

Panoramic report on 72 anti-tumor targeted drugs

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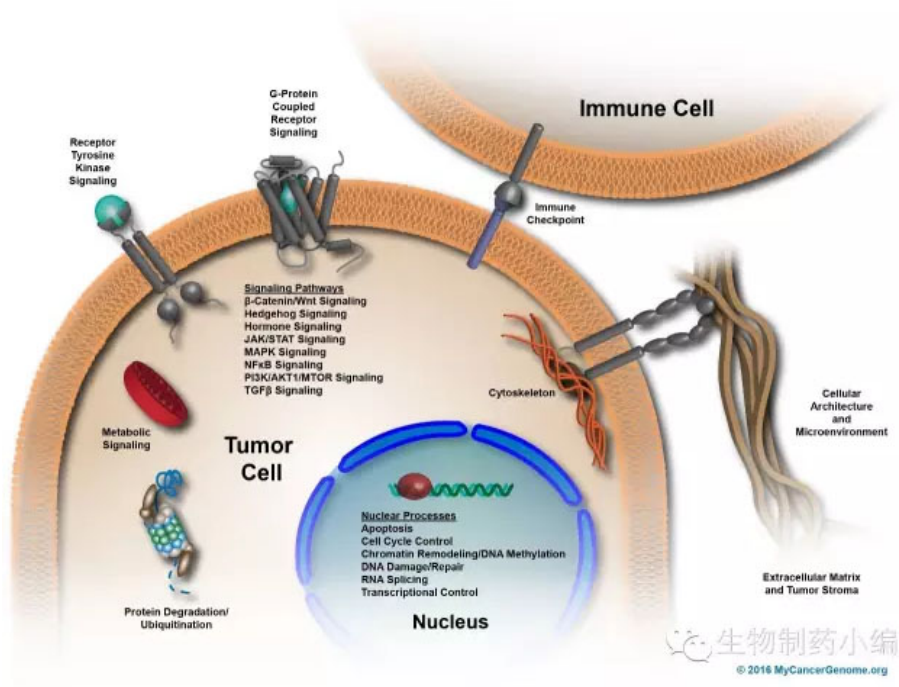
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Editor's Note

Targeted drugs have greatly changed the treatment of tumors and other indications: they have few side effects, significant efficacy, and improve the quality of life of patients, while also creating huge market opportunities. In early 2016, Nature published a special article introducing 45 anti-tumor targeted drugs approved by the FDA, mainly kinase inhibitors and antibody drugs. This article will sort out anti-tumor targeted drugs from a wider range, including a more complete range of targeted anti-tumor drugs, supplementary market structure, signal pathway introduction and other information, to paint a more complete panorama.



1. Statistics of targeted drugs

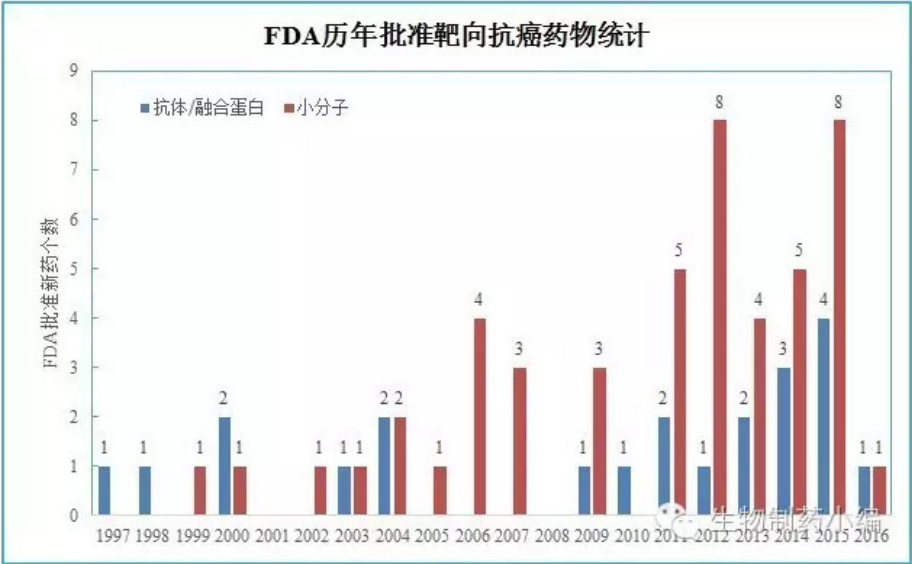
As of June 2016, the FDA has approved 72 targeted anti-tumor drugs.

附表一 2015年抗肿瘤靶向药物销售排行榜

序号	商品名	通用名	靶点	上市	企业	类别	销售额/亿美元	
							2014	2015
1	Rituxan	利妥昔单抗	CD20	1997	Roche	单抗	69.00	70.45
2	Avastin	贝伐珠单抗	VEGFR	2004	Roche	单抗	64.17	66.84
3	Herceptin	曲妥珠单抗	HER2	1998	Roche	单抗	62.75	65.38
4	Gleevec	伊马替尼	Bcr-Abl	2000	Novartis	小分子	47.46	46.58
5	Xgeva/Prolia	狄诺塞麦	RANKL	2010	Amgen	单抗	22.51	27.17
6	Erbix	西妥昔单抗	EGFR	2004	Lilly	单抗	22.97	19.96
7	Tasigna	尼罗替尼	Bcr-Abl	2007	Novartis	小分子	15.29	16.32
8	Sprycel	达沙替尼	Bcr-Abl	2006	BMS	小分子	14.93	16.20
9	Afinitor	依维莫司	mTOR	2009	Novartis	小分子	15.75	16.07
10	Perjeta	帕妥珠单抗	HER2	2012	Roche	单抗	9.18	14.45
11	Tarceva	厄洛替尼	EGFR/HER2	2004	Roche	小分子	12.92	11.81
12	Yervoy	伊匹单抗	CTLA-4	2011	BMS	单抗	13.08	11.26
13	Sutent	舒尼替尼	PDGF/VEGF	2006	Pfizer	小分子	11.74	11.20
14	Imbruvica	依鲁替尼	BTK	2013	J&J/Abbvie	小分子	2	10.32
15	Nexavar	索拉非尼	VEGF/PDGF	2005	Bayer	小分子	10.27	10.03
16	Opdivo	纳武单抗	PD-1	2014	BMS	单抗	0.05	9.42
17	Xalkori	克唑替尼	ALK	2011	Pfizer	小分子	4.38	7.80
18	Kadcyla		HER2	2013	Roche	ADC	5.36	7.69
19	Ibrance		CDK4/6	2015	Pfizer	小分子		7.23
20	Keytruda	派姆单抗	PD-1	2014	MSD	单抗	0.55	5.66
21	Votrient	帕唑帕尼	VEGF/PDGF	2009	Novartis	小分子	5.45	5.65
22	Vectibix	帕尼单抗	EGFR	2006	Amgen	单抗	5.05	5.49
23	Iressa	吉非替尼	EGFR	2003	AZ	小分子	6.23	5.43
24	Kypriolis	卡非佐米	26S proteasome	2012	Amgen	小分子	3.31	5.12
25	Xeljanz	托法替尼	JAK	2011	Pfizer	小分子	3.08	5.23
26	Adcetris		CD30	2011	Takeda	ADC	3.70	4.68
27	Inlyta	阿昔替尼	VEGFR	2012	Pfizer	小分子	4.10	4.30
28	Jakafi		JAK1/2	2011	Novartis	小分子	2.79	4.10
29	Cyramza	雷莫芦单抗	VEGFR2	2014	Lilly	单抗	0.76	3.84
30	Stivarga	瑞戈非尼	多激酶抑制剂	2012	Bayer	小分子	2.98	3.52
31	Campath	阿伦单抗	CD52	2000	Millenium	单抗	0.45	2.73
32	Zelboraf	威罗非尼	B-Raf	2011	Roche	小分子	3.01	2.14
33	Rapamune	西罗莫司	mTOR	1999	Pfizer	小分子	3.39	1.97
34	Zydelig		PI3K-δ	2014	Gilead	小分子	0.23	1.32
35	Iclusig	泊那替尼	Bcr-Abl	2012	Ariad	小分子	0.56	1.13
36	Lenvima	乐伐替尼	RTK	2015	Esai	小分子	0.03	1
37	Lynparza	奥拉帕尼	PARP	2014	AZ	小分子		0.94
38	Zaltrap	阿柏西普	VEGFR	2011	Sanofi	融合	0.78	0.87
39	Zykadia	色瑞替尼	ALK	2014	Novartis	小分子	0.31	0.79
40	Istodax	罗米地辛	HDAC	2009	Celgene	小分子	0.66	0.69
41	Tagrisso		EGFR	2015	AZ	小分子		0.19
42	Zevalin	替伊莫单抗	CD20	2002	Spectrum	单抗	0.22	0.18
43	Empliciti		SLAMF7	2015	Abbvie	单抗		0.03
44	Alecensa	艾乐替尼	ALK	2015	Roche	小分子		
45	Arzerra	奥法本单抗	CD20	2009	GSK	单抗	0.72	
46	Beleodaq	贝利司他	HDAC	2014	Spectrum	小分子		
47	Bexxar	托西莫单抗	CD20	2003	Smithkline	单抗	退市	
48	Bosulif	博舒替尼	Src/Abl	2012	Pfizer	小分子		
49	Caprelsa	凡德他尼	多激酶抑制剂	2011	Sanofi	小分子		
50	Cometriq	卡博替尼	C-Met	2012	Exelixis	小分子		
51	Cotellic		MEK	2015	Roche	小分子		
52	Dazalex		CD38	2015	J&J	单抗		
53	Erivedge		Hedgehog	2012	Roche	小分子		
54	Farydak	帕比司他	HDAC	2015	Novartis	小分子		
55	Gazyva		CD20	2013	Roche	单抗	0.49	
56	Gilotrif	阿法替尼	EGFR	2013	BI	小分子		
57	Kepivance	帕利夫明	FGFR	2004	Amgen	小分子		
58	Mekinist	曲美替尼	MEK1/2	2013	GSK	小分子		
59	Mylotarg	吉姆单抗	CD33	2000	Pfizer	单抗		
60	Ninlaro		20S proteasome	2015	Takeda	小分子		
61	Odomzo		Hedgehog	2015	Novartis	小分子		
62	Portrazza		EGFR	2015	Lilly	小分子		
63	Synribo		Bcr-Abl	2012	Teva	小分子		
64	Tafinlar	达拉非尼	BRAF	2013	GSK	小分子		

65	Torisel	坦西莫司	mTOR	2007	PF Prism CV	小分子		
66	Tykerb	拉帕替尼	EGFR/erb2	2007	GSK	小分子		
67	Unituxin		GD2	2015	United	单抗		
68	Zolinza	伏立诺他	HDAC	2006	MSD	小分子		
69	Blinicyto		BiTE	2014	Amgen	双抗	0.03	
70	Portrazza	Necitumumab	BAFF	2015	Lilly	单抗		
71	Tecentriq	Atezolizumab	PD-L1	2016	Roche	单抗		
72	Venclexta	Venetoclax	BCL-2	2016	Abbvie	小分子		
	Provenge	Sipuleucel-T		2010	Valeant	疫苗		2.59
合计							448.83	507.07

In recent years, targeted anti-cancer drugs, both monoclonal antibodies and small molecule targeted drugs, have shown a rapid growth trend.



Among targeted anti-cancer drugs, monoclonal antibodies account for 60% of the market share and small molecule drugs account for 40% of the market share.



The market concentration of targeted anti-cancer drugs is very high, with the top five pharmaceutical companies: Roche, Novartis, Amgen, BMS, and Pfizer accounting for more than 85% of the market share. Roche alone accounts for nearly half of the market, and the combined sales of the three major products: rituximab, herceptin, and bevacizumab exceed US\$20 billion.



The forms of new drugs are becoming more and more diverse, such as CAR-T, CAR-NK, bispecific antibodies, Aptamer, etc., which will bring more possibilities to tumor targeted therapy. I will not go into details here.

For more information, see the previous article. [Click the title to open the original text:](#)

1. [Anti-tumor drug research report](#)
2. [2015 Global Prescription Drug Market Research Report: Biologics Rise Steadily](#)
3. [Panoramic report of 61 marketed antibody drugs](#)
4. [Antibody/fusion protein drug research report](#)
5. [Statistics of 114 bispecific antibody drugs under development](#)

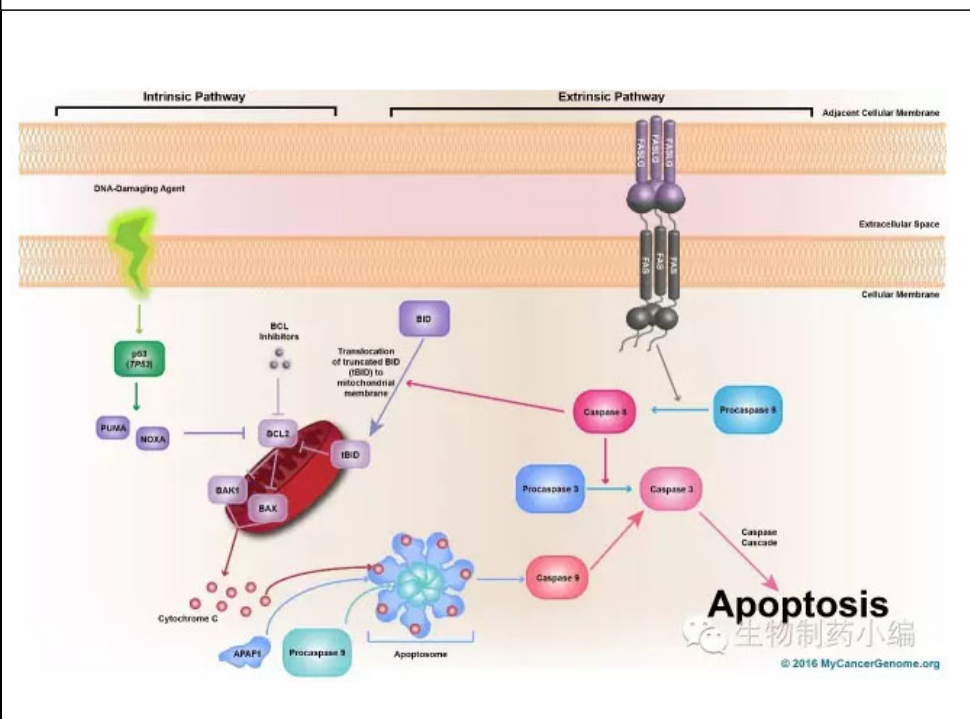
2. Inventory of signal pathways

Refer to My Cancer Genome and sort out the relevant signal pathway information as follows.

1. Apoptosis pathway: programmed cell death process. The extracellular part is mediated by **FAS** ligand /**FAS** receptor, and the intracellular part is mediated by mitochondria.

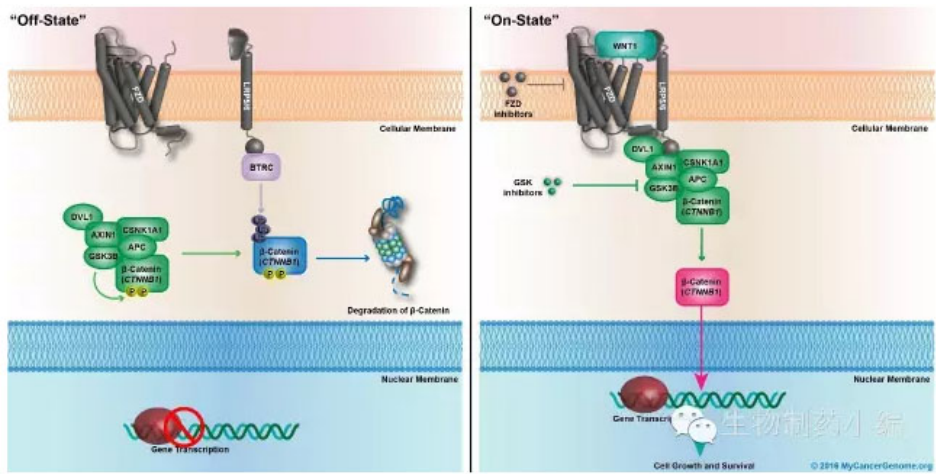
Drug target: **BCL-2**

Representative drug: **Venclexta**



2. β -Catenin/WNT pathway: It activates gene transcription to achieve multiple effects including cell growth. **WNT1** ligand binds to **FZD1** (a **GPCR**) to activate this pathway.

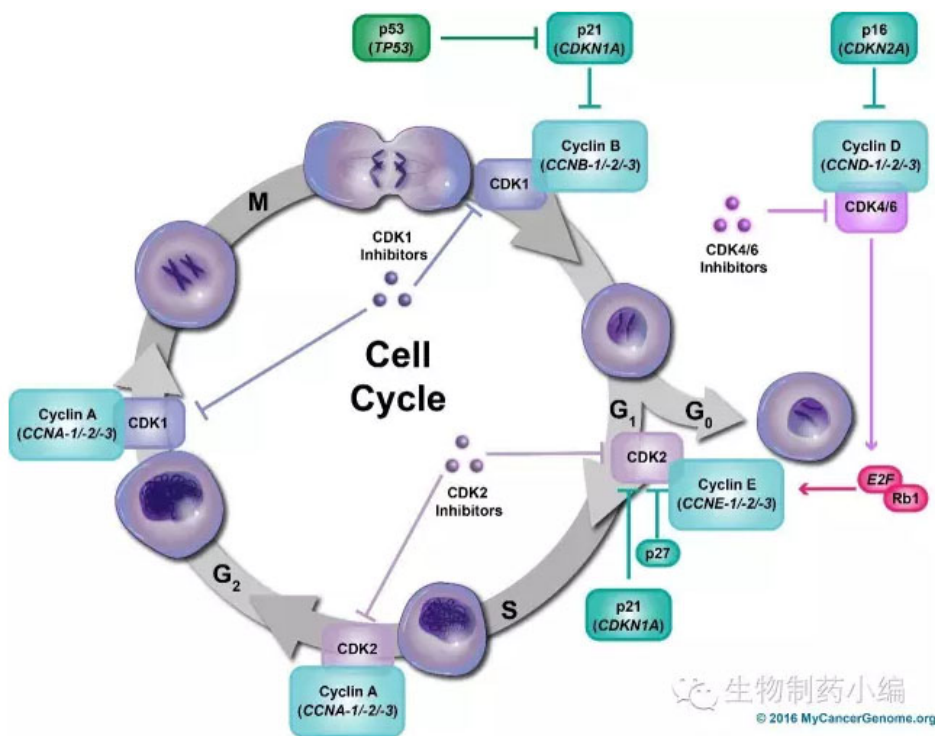
Drug target: **FZD**, etc.



3. Cell cycle control pathway: The cell cycle includes **G1**, **S**, **G2**, and **M** phases, and cycle regulation depends on a series of cycle checkpoints: **CDK4/6+Cyclin D**, **RB1+E2F**, **CDK2+Cyclin E**, **CDK2+Cyclin A**, **CDK1+Cyclin A**, **CDK1+Cyclin B**.

Drug target: **CDK4/6**

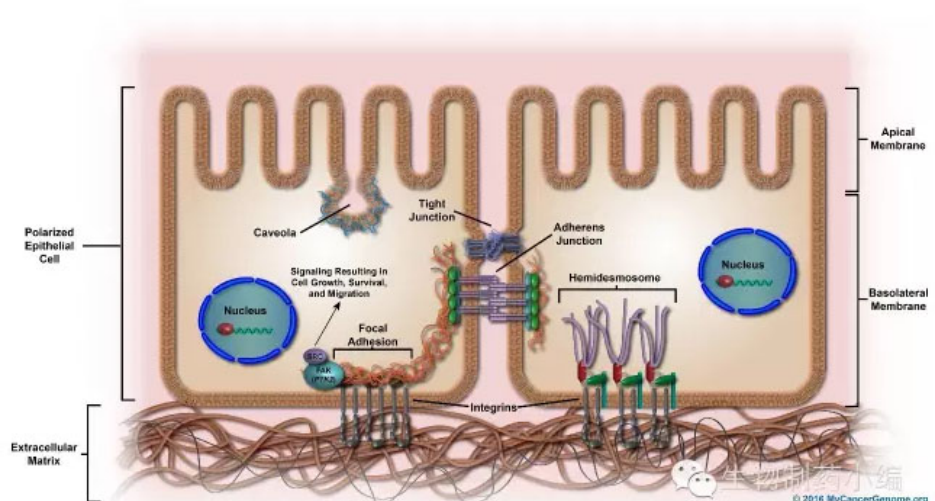
Representative drug: **Ibrance**



4. Cellular structure and microenvironment: The intracellular and intercellular structures of tumor cells are very important. For example, chemical regulatory molecules can be transmitted within or between cells through these structures.

Drug target: **SRC**

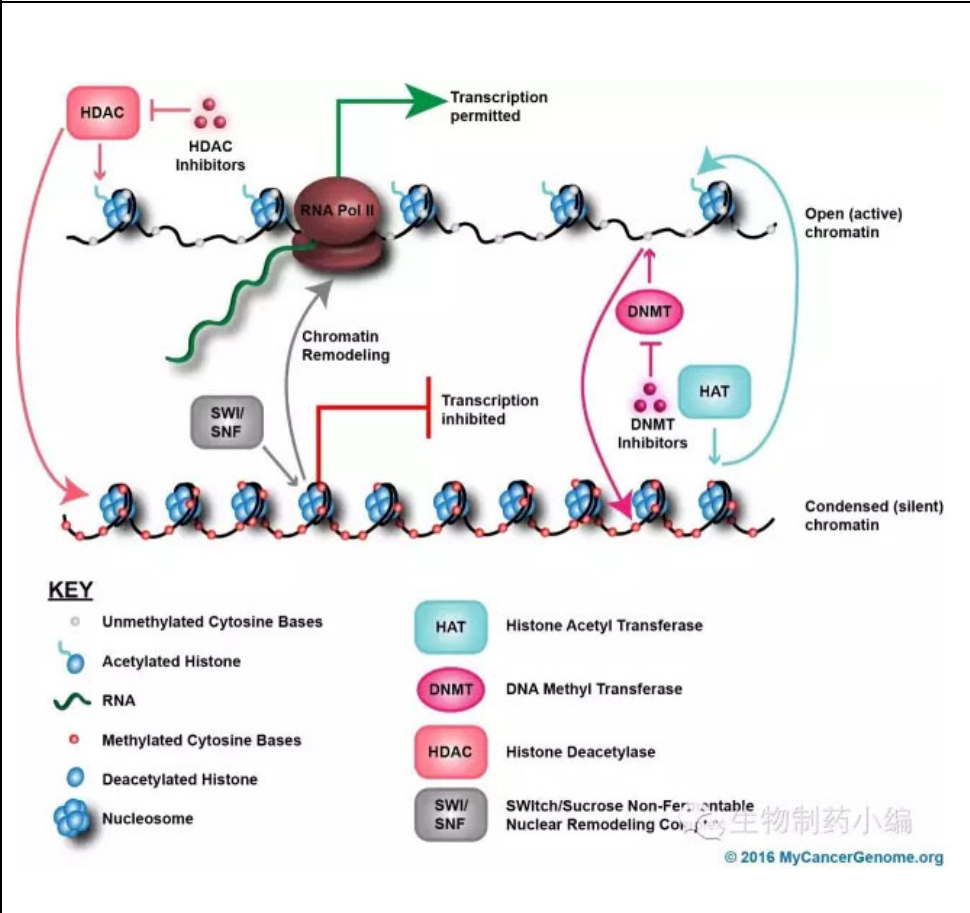
Representative drug: **Sprycel** (also targets **ABL**)



5. Chromatin remodeling /DNA methylation: Chromatin remodeling can make the originally highly compressed DNA structure transcribed. The remodeling process is generally regulated by histone modifications such as DNA methylation (an epigenetic regulatory mechanism).

Drug Target: HDAC

Representative drugs: Istodax, Farydak, Beleodaq, Zolniza, Chidamide



6. Cytotoxic chemotherapy drugs: usually refer to substances that cause cell death, killing cells by inhibiting microtubules, protein function, and DNA synthesis.

Drug targets: alkylating agents, metabolic antagonists, tubulin inhibitors, topoisomerase II inhibitors, etc.

Representative drugs: nitrogen mustard, methotrexate, paclitaxel, etoposide

M Phase Specific**Antimicrotubule Agents**

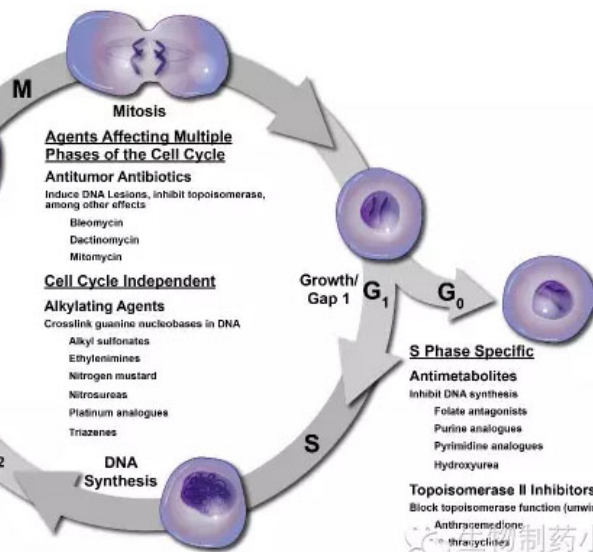
Inhibit function of microtubules

- Epothilones
- Halichondrin B analogue
- Taxanes
- Vinca alkaloids

Topoisomerase II Inhibitors

Block topoisomerase function (unwinding DNA)

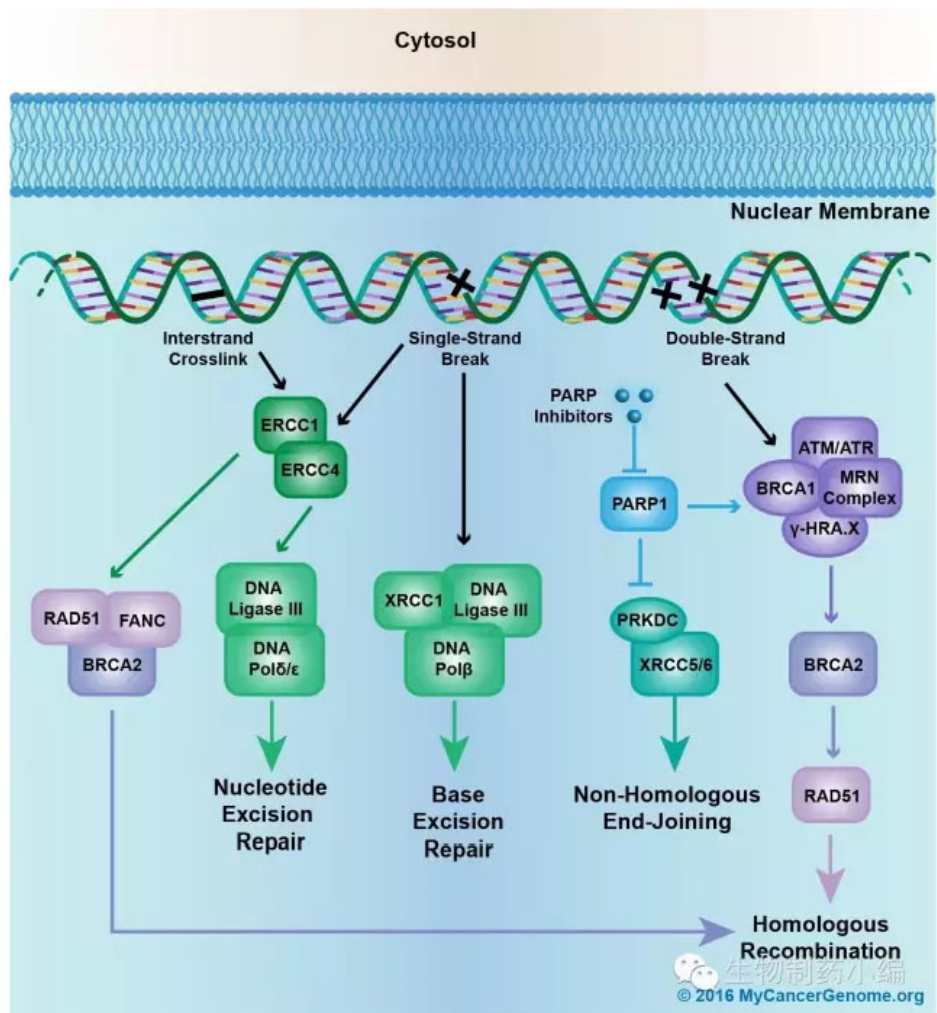
- Anthrasedione
- Anthracyclines
- Epidodophyllotoxins



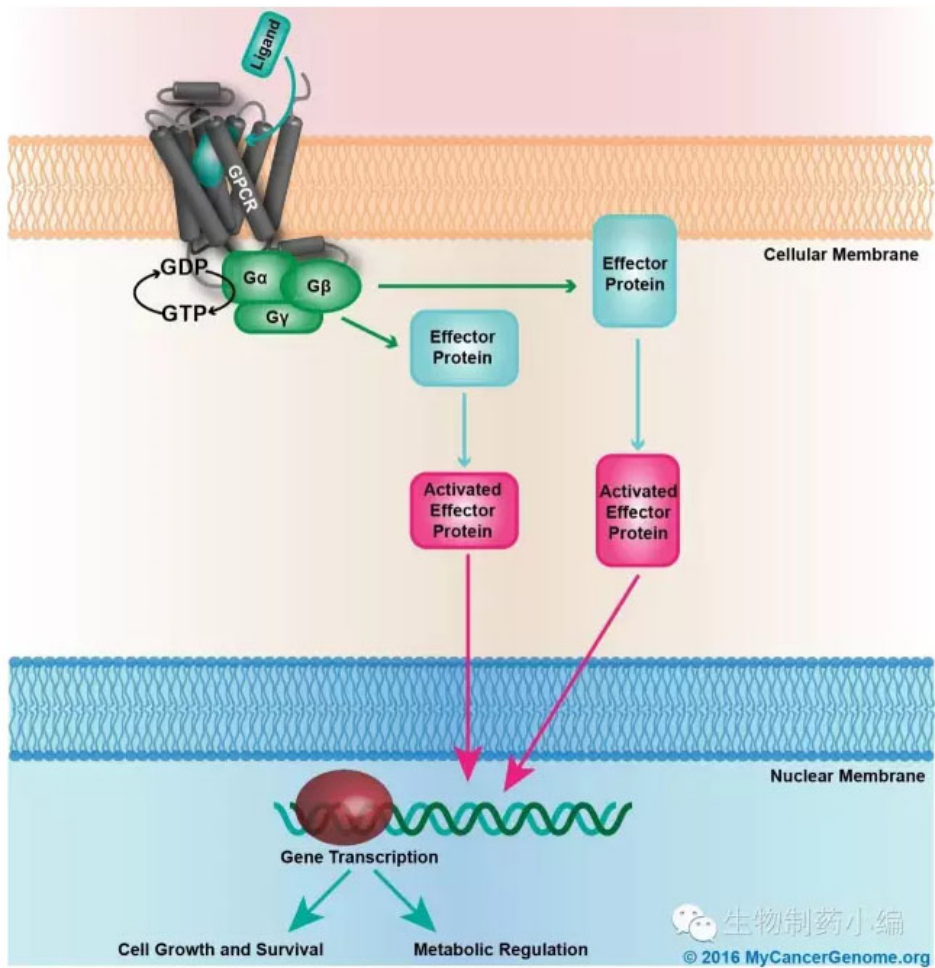
7. **DNA damage /repair:** Internal factors such as oxygen free radicals, replication errors, oxidative deamination and external factors such as ultraviolet radiation can cause **DNA damage**. If the **DNA repair** pathway is blocked, cells will enter aging or programmed death.

Drug Target: **PARP**

Representative drugs: **Lynparza** , **Niraparib** , **Talazoparib** , **Rucaparib** , **BGB-290** , etc.



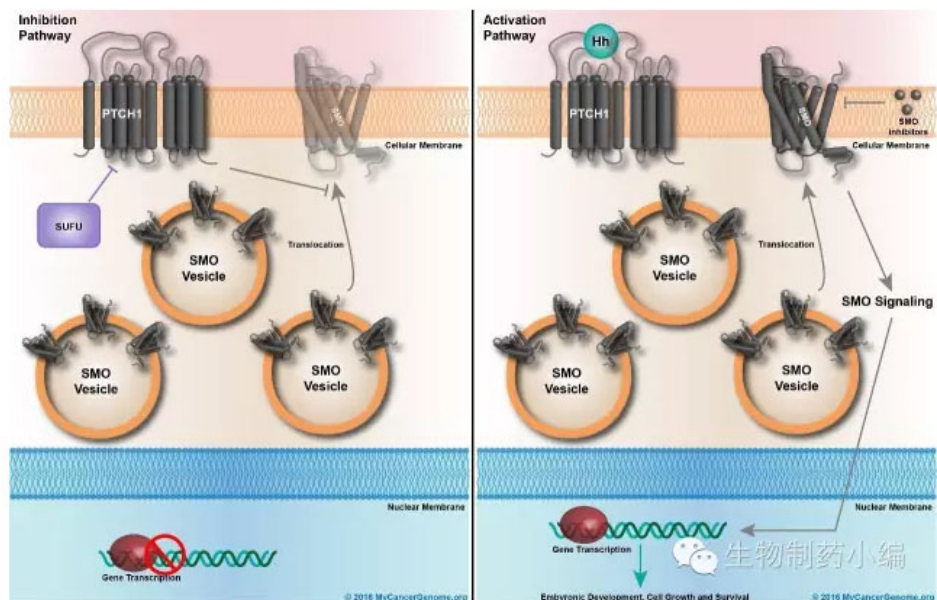
8. **G-蛋白信号通路**：该通路在代谢调控、神经传导、胚胎发育中发挥作用。**G**蛋白信号通路通过配基结合**GPCR**激活，蛋白激酶结合**GPCR**可导致后者的磷酸化，进而抑制该信号通路。



9. **Hedgehog**信号通路：该通路促进胚胎发育过程中的细胞生长和分化，在癌细胞中通常会异常活化。**Hedgehog**配基结合细胞表面的**PTCH1**，激活该通路。在配基缺失的情况下，通路被**PTCH1**抑制，通过把**SMO**锁定在小泡结构中抑制**SMO**的活化。

药物靶点：**SMO**

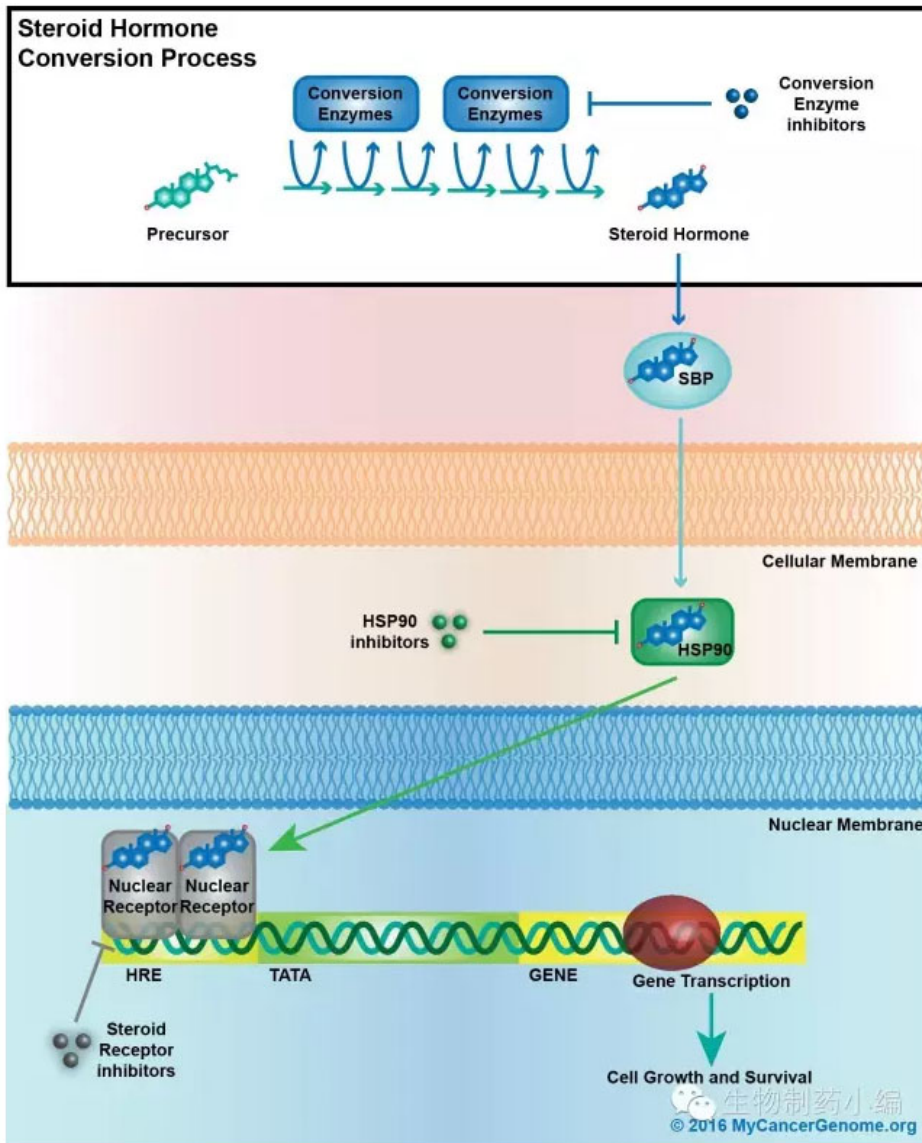
代表药物：**Erivedge**



10. 激素信号通路：类固醇激素信号通路在激活转录和基因表达过程中起作用，通路可被类固醇激素如雌激素、孕激素激活。

药物靶点：芳香酶抑制剂、HSP90抑制剂、类固醇受体（雄激素、雌激素、孕激素）

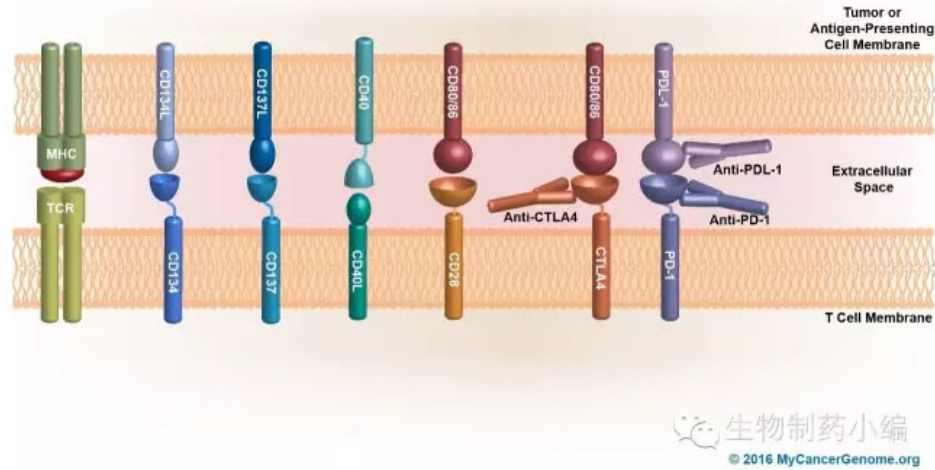
代表药物：Femara、Luminespib、Casodex、Faslodax、他莫昔芬



11. 免疫检验点：如CTLA4在CD4+和CD8+的T细胞表达，PD-1在T细胞和B细胞表达，两者都可以组织T细胞的活化。通过阻断该通路可以激活T细胞，杀死肿瘤细胞。后来又出现一些需激活的免疫检验点。

药物靶点：CTLA4-Fc、PD-1、PD-L1、OX-40等

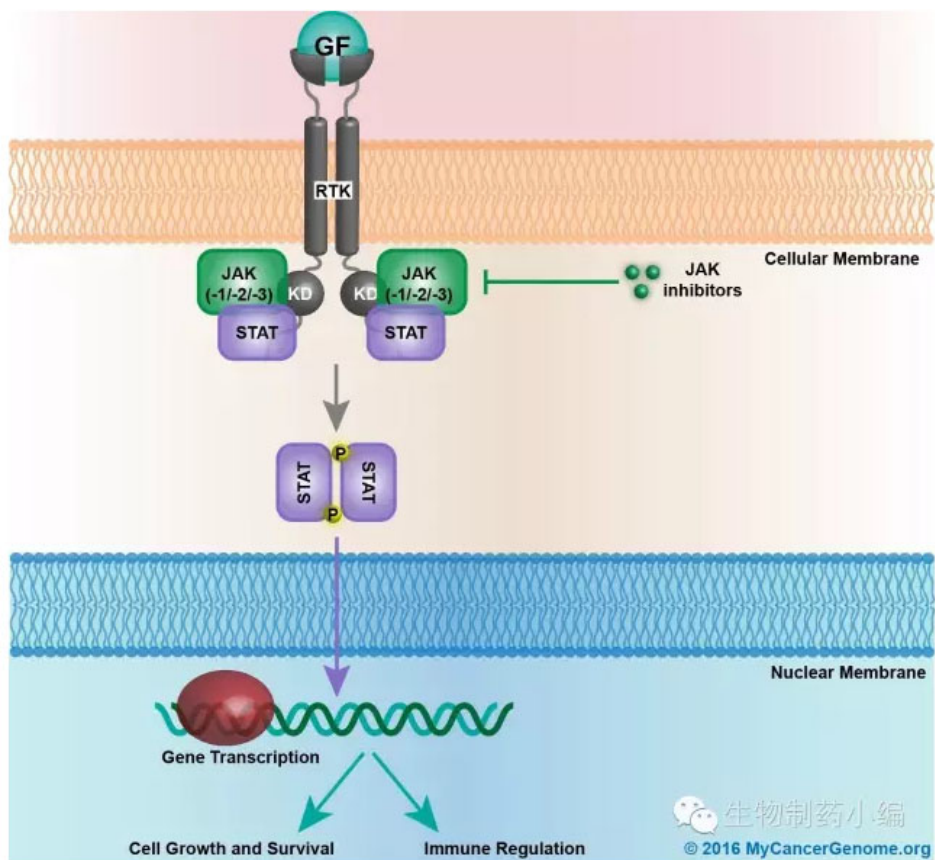
代表药物：Yervoy、Keytruda、Opdivo、Tecentriq



12. JAK-STAT信号通路：JAK-STAT信号通路在基因转录和免疫调节中发挥作用。**JAK1/2/3**是胞内的酪氨酸激酶，**STAT**则是激活基因表达的转录因子。该通路通过结合配基如白介素、干扰素、生长因子等激活。

药物靶点：**JAK**

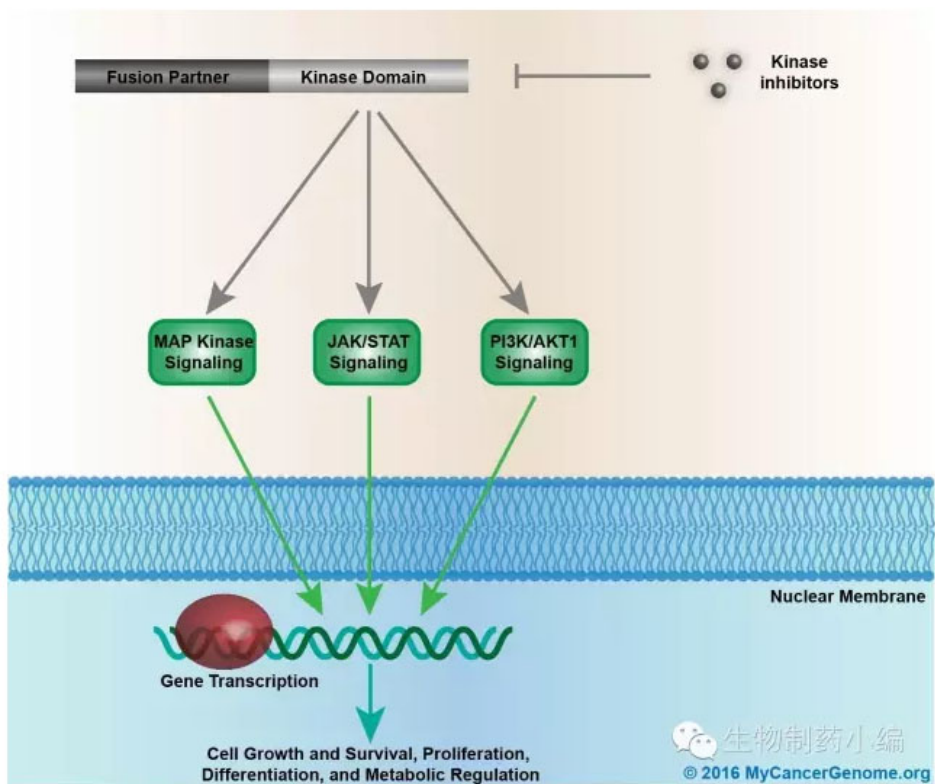
代表药物：**Xeljanz、Jakafi**



13. 激酶融合通路：激酶融合是肿瘤细胞中发生的2个基因重排的现象，其中1个是激酶基因。激酶融合促进基因转录、细胞生长、分化和存活。

药物靶点：JAK、ABL、ALK、FGFE1/2/3、NTRK1/2/3、RET、ROS1等

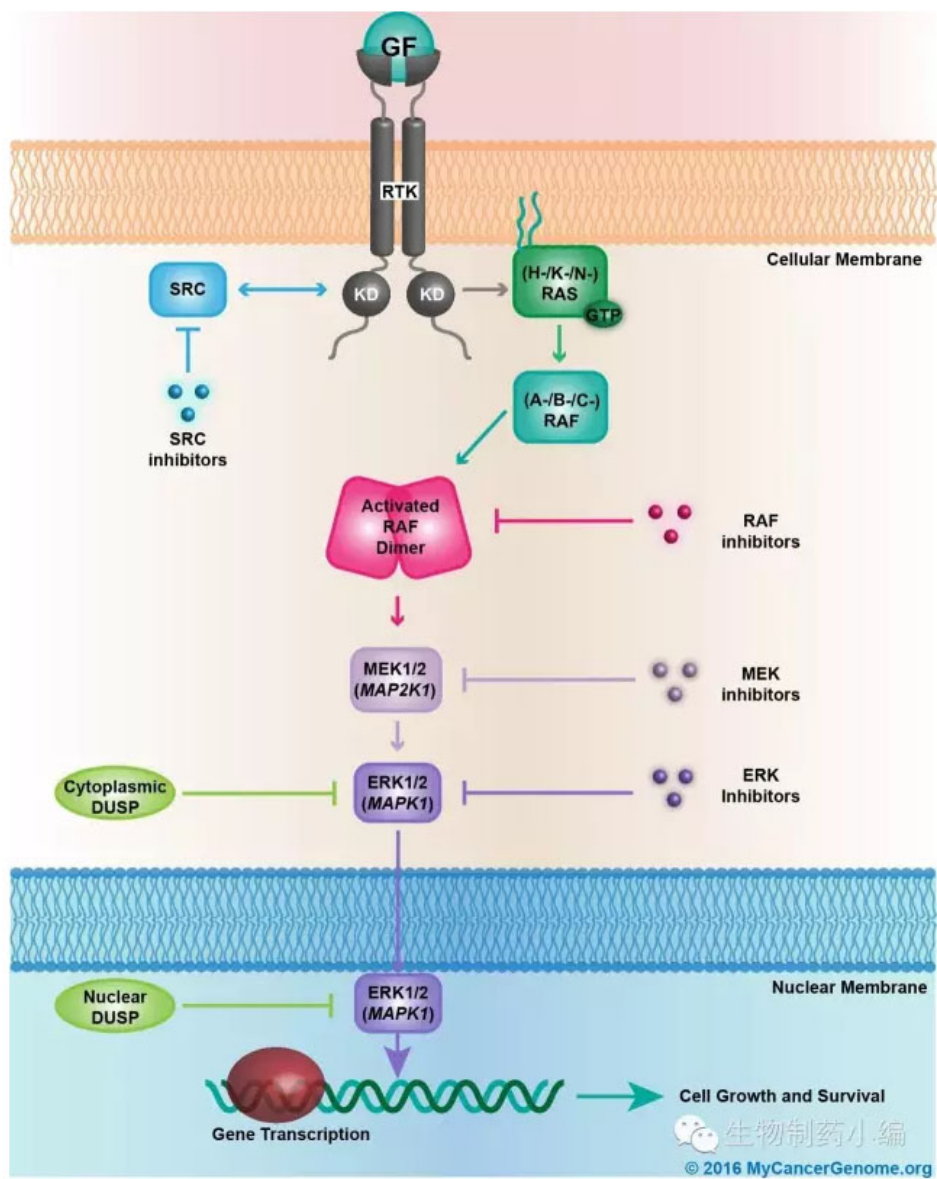
代表药物：Jakafi、Sprycel、Alecensa、Iclusig、ENTrectinib、Nexavar等



14. MAPK通路：分裂素活化蛋白激酶信号通路促进细胞生长、分化和存活。

药物靶点：BRAF、ERK、MEK、SRC

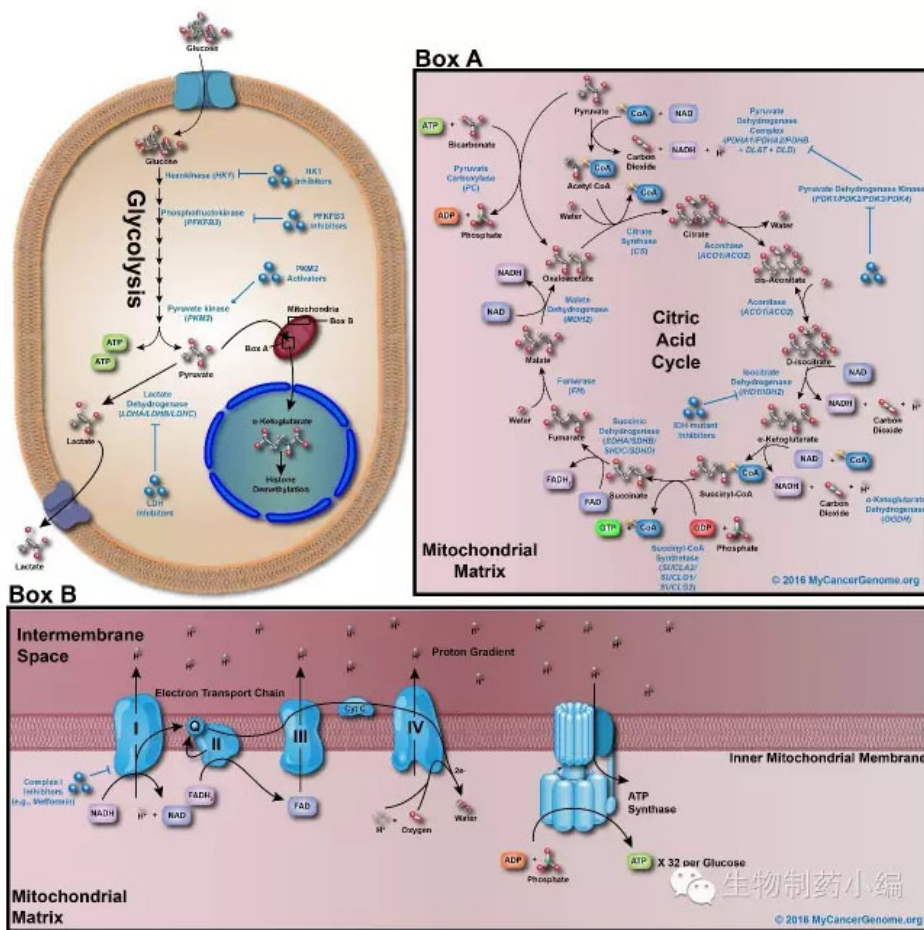
代表药物：Zelboraf、Ralimetinib、Mekinist、Saracatinib



15. 代谢通路：代谢通路是指通过糖有氧呼吸和无氧呼吸的分解产生能量，促进细胞生长、分化和存活。

药物靶点：IDH1

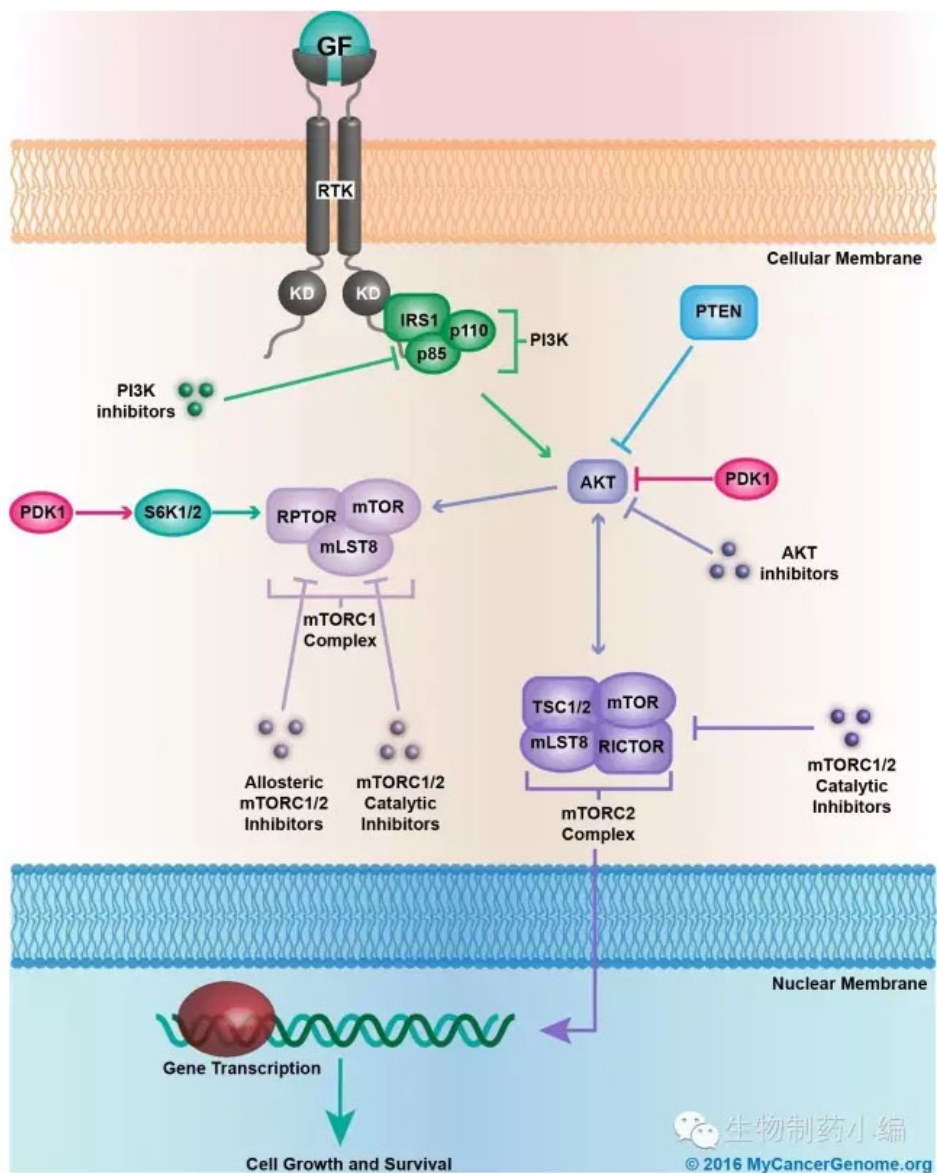
代表药物：AG-120



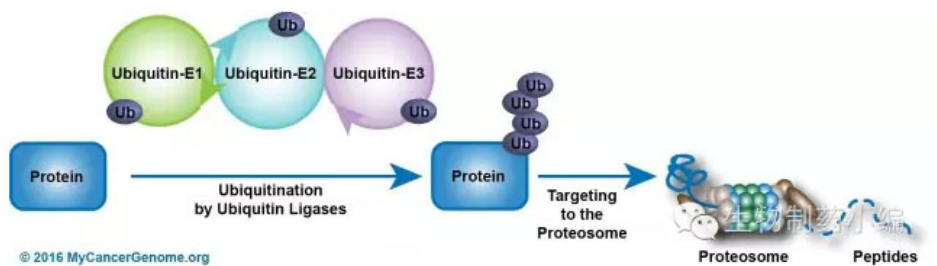
16. PI3K/AKT1/mTOR通路：该通路在细胞生长、蛋白翻译、凋亡的调控中发挥作用。该通路通过保外生长因子如IGF1结合相应的RTK来激活，同时可被PTEN抑制。

药物靶点：mTOR

代表药物：Affinitor、Torisel



17. Protein degradation /ubiquitination: Ubiquitination modifies substrate proteins and mediates their degradation by the proteasome. Ubiquitination can also affect protein activity and interactions.

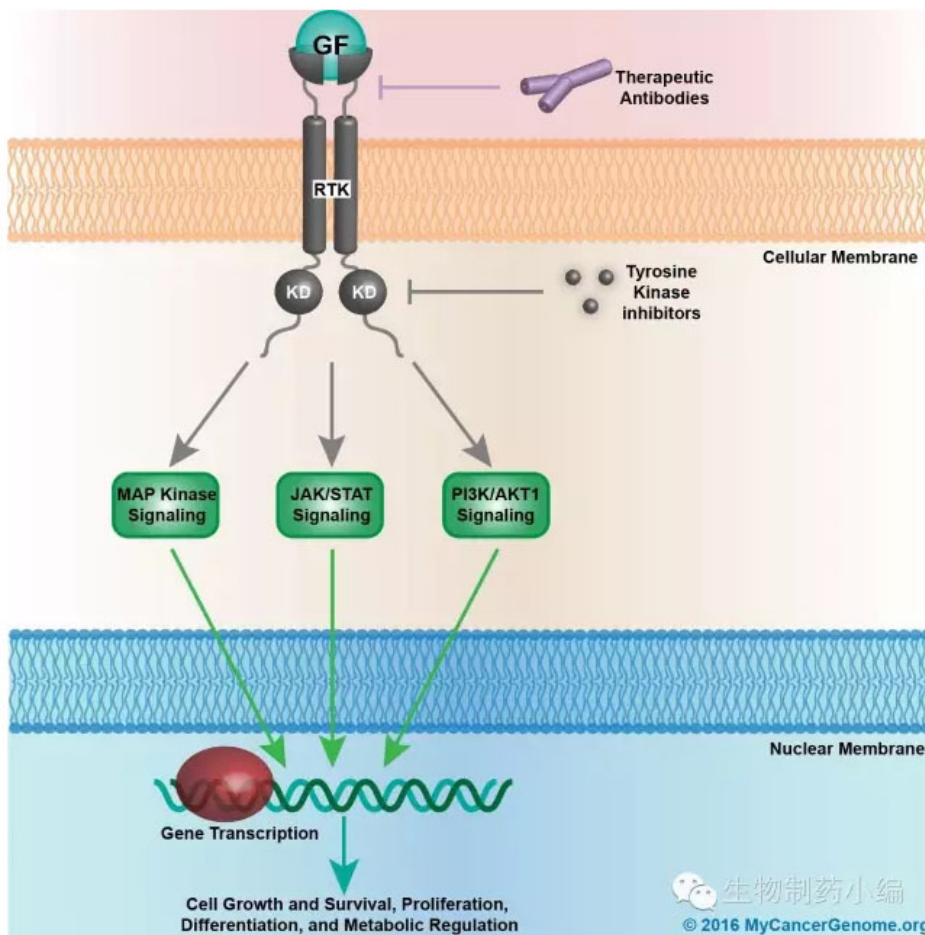


18. RTK/GF signaling pathway: The receptor tyrosine k

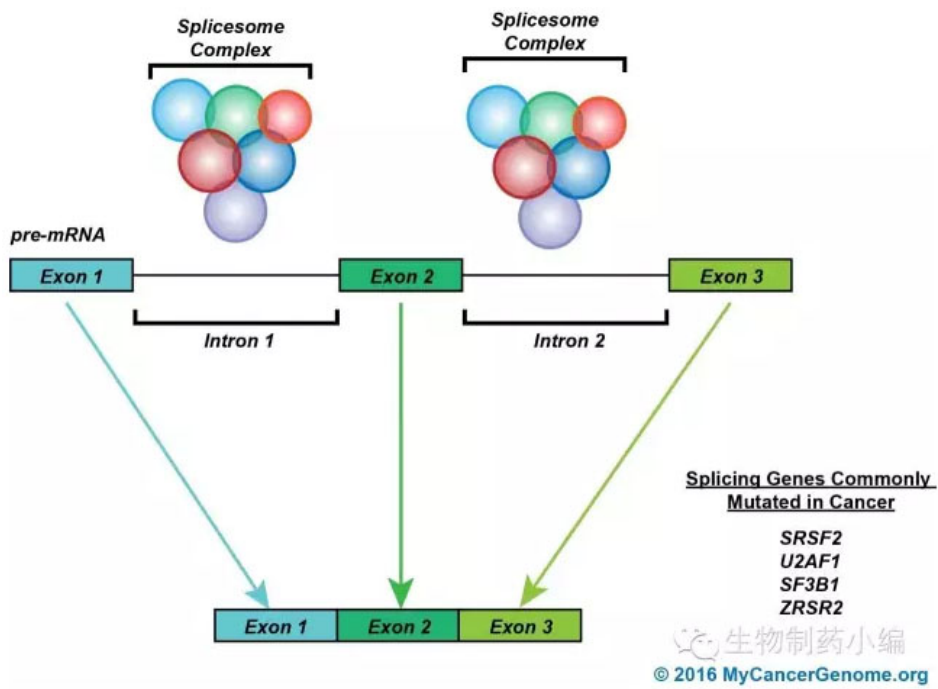
inase /growth factor signaling pathway plays a role in cell growth, differentiation, survival, gene transcription, metabolic regulation, etc. This pathway is activated by the binding of ligands such as growth factors.

Drug target: tyrosine kinase **TKI**

Representative drugs: **Sprycel**, **Gleevec**, etc.



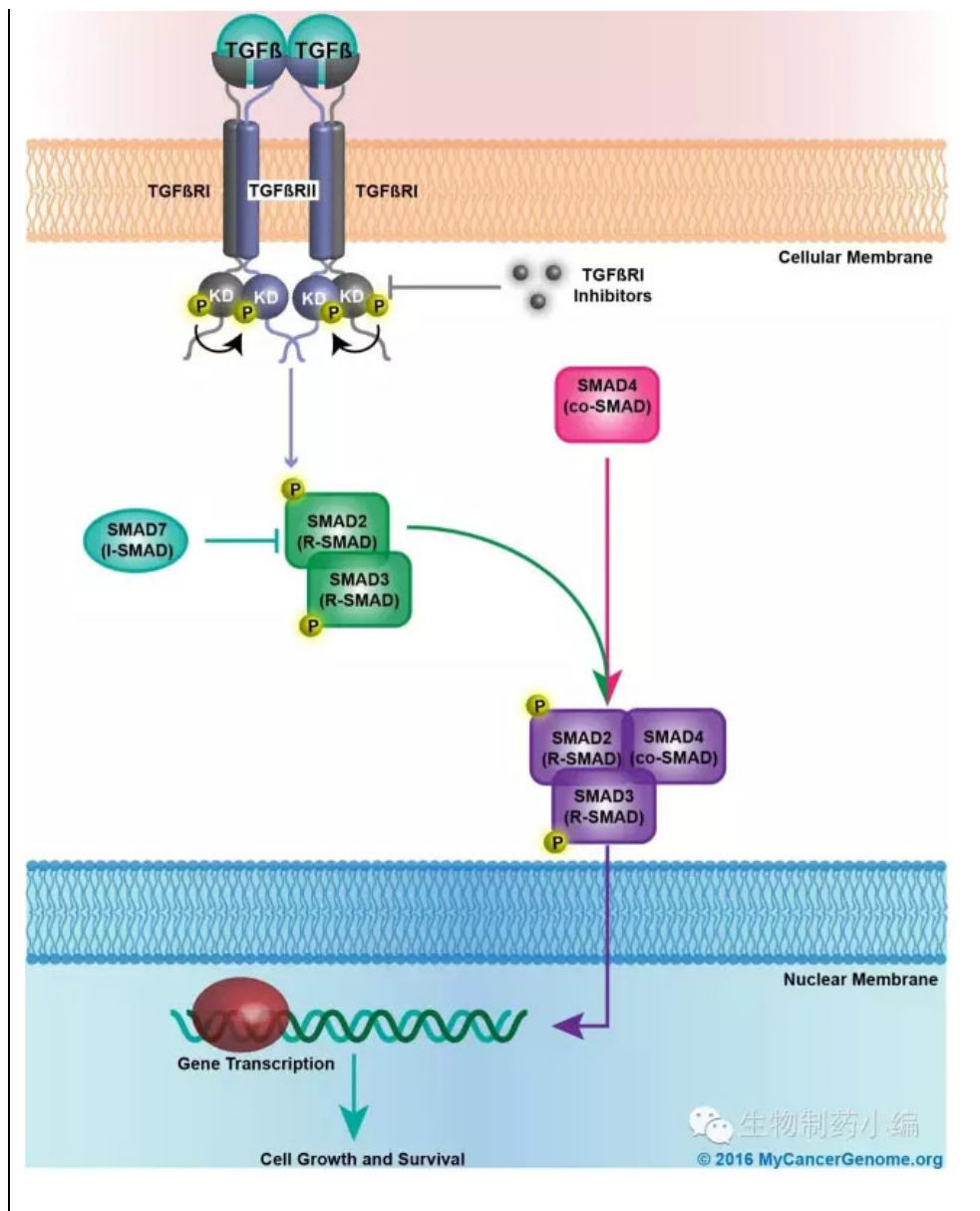
19. RNA splicing: RNA splicing refers to the process of intron splicing and exon ligation, which is activated by the binding of RNA molecules to the spliceosome.



20. **TGF β** pathway: **TGF β** pathway plays an important role in the occurrence, progression and metastasis of cancer. **SMAD** plays an important role in signal transduction.

Drug target: **TGFR1**

Representative drug: **Galunisertib**



3. Statistics of anti-tumor drug targets

Appendix 2 Statistics of anti-tumor drug targets

激酶抑制剂				
靶点	研发代码	通用名	商品名	备注
ABL	AMN107	nilotinib	Tasigna	
	AP24534	ponatinib	Iclusig	
	AS703569, R763	cenisertib		
	AT9283			
	BMS-354825	dasatinib	Sprycel	
	INNO-406, NS-187	baftatinib		
	SKI-606	bosutinib	Bosulif	
	STI-571, CGP57148, CGP57148B	imatinib	Gleevec	
AKT	XL228			
	AS703569, R763	cenisertib		catalytic inhibitor
	AT13148			
	AZD5363			catalytic inhibitor
	BAY1125976			
	GDC-0068	ipatasertib		catalytic inhibitor
	GSK2110183	afuresertib		
	GSK2141795	uprosertib		
ALK (Anaplastic Lymphoma Kinase)	MK-2206			allosteric inhibitor
	ACE-041	dalantercept		
	AP26113	brigatinib		
	ASP2215	gilteritinib		
	ASP3026			
	CH5424802, RO5424802, AF802	alectinib	Alecensa	
	LDK378	ceritinib	Zykadia	
	PF-02341066	crizotinib	Xalkori	
	PF-06463922	lorlatinib		
	RXDX-101	entrectinib		
ATR	TSR-011			
	X-396			
Aurora A	VX-803			
	AS703569, R763	cenisertib		
	AT9283			
	ENMD-2076			
	MK5108			
	MLN8237	alisertib		
	PF-03814735			
	TAK901			
Aurora B	TAS-119			
	AS703569, R763	cenisertib		
	AT9283			
	AZD1152-HQPA, AZD1152	barasertib		
Aurora C	GSK1070916			
	PF-03814735			
Aurora A/B/C	GSK1070916			
	ABT-348	ilorasertib		
	AMG900			
	BI 847325			
	PHA-739358	danusertib		
AXL	SNS-314			
	ASP2215	gilteritinib		
	MGCD265			
BRAF	MGCD516			
	ARQ 736			
	BGB-283			
	GSK2118436	dabrafenib	Tafinlar	
	PLX-4032, RG7204, R05185426	vemurafenib	Zelboraf	
	RAF265			
BTK	RO5212054, PLX3603			
	AS703569, R763	cenisertib		
	AVL-292			
CDK 1,2,4,5,6,7,9 (pan-CDK)	PCI-32765	ibrutinib	Imbruvica	
	HMR-1275, L868275, flavopiridol	alvocidib		
	CDK 1,2,4,7,9 (pan-CDK)	BAY 1000349	roniciclib	
	CDK 1,2,5,9	SCH727965	dinaciclib	
CDK 2,7,9	CYC202	seliciclib		

CDK 4,6	LEE011	ribociclib		
	LY2835219	abemaciclib		
	PD 0332991	palbociclib	Ibrance	
CDK 9/PTEFb (positive transcription elongation factor b)	BAY1143572			
CHEK1	LY2606368	prexasertib		
CHEK2	LY2603618	rabusertib		
CSF1R	PLX3397, PLX108-01			
	PLX7486			
DDR2	MGCD516			

生物制药小编

EGFR (HER1/ERB B1)	AEE788, NVP- AEE788			
	AP26113	brigatinib		wild type-sparing EGFR inhibitor
	ASP8273	naquotinib		wild type-sparing EGFR inhibitor
	AZD6474, ZD6474	vandetanib	Zactima, Caprelsa	
	AZD9291	osimertinib	Tagrisso	wild type-sparing EGFR inhibitor
	BGB-283			
	BIBW-2992	afatinib	Gilotrif, Tomtovok	
	BPI-2009H	icotinib		
	CI-1033, PD183805	canertinib		
	CO-1686	rociletinib		wild type-sparing EGFR inhibitor
	EGF816			wild type-sparing EGFR inhibitor
	HM61713			wild type-sparing EGFR inhibitor
	HMPL-813	epitinib		
	MNPL-309	theliatinib		
	OSI-774, CP-258,774	erlotinib	Tarceva	
EGFR (HER1/ERB B1) HER2 (ERBB2/ne u) HER3 (ERBB3)	XL647, KD019			
	ZD1839	gefitinib	Iressa	
EGFR (HER1/ERB B1) HER2 (ERBB2/ne u) HER4 (ERBB4)	AZD8931			
	BMS-599626, AC480			
	BMS-690514, EVR1			
	EKB-569	pelitinib		
EphB4	HM781-36B	poziotinib		
	PF-00299804	dacomitinib		
FAK	MGCD516			
	XL647, KD019			
	AB1010	masitinib		
	GSK2256098			
FGFR1/2/3	VS-4718			
	VS-6063	defactinib		
	AB1010	masitinib		
	AP24534	ponatinib	Iclusig	
	AZD4547			
	BGJ398, NVP- BGJ398			
	BIBF1120	nintedanib	Vargatef	
	Debio 1347, CHS183284			
	E-3810	lucitanib		
	HMPL-012	sulfatinib		
FGFR1/2/3/ 4	LY2874455			
	TKI258, CHIR-258	dovitinib		
FGFR1	XL228			
FGFR3	JNJ-42756493			
FLT3	TSU 68, SU6668	orantinib		
	ENMD-2076			
	AC220	quizartinib		
	AP24534	ponatinib	Iclusig	
	AS703569, R763	cenisertib		
	ASP2215	gilteritinib		
	BAY 43-9006	sorafenib	Nexavar	
	CEP-701	lestaurtinib		
	CP-868569	crenolanib		
	ENMD-0276			
	MLN518, CT53518	tandutinib		
	MP-470	amuvatinib		
	PKC412	midostaurin		
	PLX3397, PLX108- 01			
HER2 (ERBB2/ne u)	SU-11248	sunitinib	Sutent	
	XL184	cabozantinib	Cabometyx (tablet), Cometriq (capsule)	
	AEE788, NVP- AEE788			
	ARRY 380			
	BIBW-2992	afatinib	Gilotrif, Tomtovok	
	CI-1033, PD183805	canertinib		
	GSK572016, GW2016, GW- 572016	lapatinib	Tykerb	
	HKI-272	neratinib		
HGFR	TAK 165	mubritinib		
	XL647, KD019			
IGF-1R	MGCD516			
	BMS-536924			
	BMS-754807			
	OSI-906	linsitinib		
JAK1	XL228			
	PRT062070			
	INCB039110	itacitinib		

JAK1/2	INCB052793			
	AZD1480			
	CYT-387, CYT387	momelotinib		
	INCB018424	ruxolitinib	Jakafi	
JAK2	AT9283			
	BMS-911543			
	CEP-701	lestaurtinib		
	SAR302503, TG101348	fedratinib		
	SB1518	pacritinib		
	XL019			
JAK2 V617F	SB1518	pacritinib		
	LY2784544	gandotinib		
	XL019			
JAK3	CP-690550	tofacitinib citrate	Xeljanz	
	PRT062070			

KIT	AB1010	masitinib		
	AG-013736	axitinib	Inlyta	
	AMG706	motesanib		
	AS703569, R763	cenisertib		
	BAY 57-9352	telatinib		
	BAY 73-4506	regorafenib	Stivarga	
	BMS-354825	dasatinib	Sprycel	
	GW786034	pazopanib	Votrient	
	MGCD516			
	MLN518, CT53518	tandutinib		
	MP-470	amuvatinib		
	PKC412	midostaurin		
	PLX3397, PLX108-01			
	STI-571, CGP57148, CGP57148B	imatinib	Gleevec	
	SU-11248	sunitinib	Sutent	
KRAS	XL184	cabozantinib	Cabometyx (tablet), Cometriq (capsule)	
	XL820			wild type-sparing KRAS inhibitor
LYN	BGB-283			
	INNO-406, NS-187	bafetinib		
MAPK1 (ERK)	AS703569, R763	cenisertib		
	BVD-523			
	LY2228820	ralimetinib		
MEK	MK-8353, SCH900353			
	AS703026, MSC1936369B	pimasertib		
	AZD6244, ARRY-142886	selumetinib		
	AZD8330			
	BAY 86-9766, RDEA119	refametinib		
	BI 847325			
	CI-1040			
	GDC-0623			
	GDC-0973	cobimetinib	Cotellic	
	GSK1120212	trametinib	Mekinist	
	MEK162	binimetinib		
	PD0325901, PD-0325901			
	RO5126766, CH5126766			
	TAK-733			
MET	AMG 208			
	AMG 337			
	ARQ 197	tivantinib		
	BMS-777607			
	EMD 1214063, MSC2156119J			
	EXEL-2880, GSK1363089, XL880	foretinib		
	HMPL-504	volitinib		
	INCB028060, INC280			
	MGCD265			
	MGCD516			
	MK-2461			
	MP-470	amuvatinib		
	PF-02341066	crizotinib	Xalkori	
	PF-04217903			
	SAR125844			
MST1R	XL184	cabozantinib	Cabometyx (tablet), Cometriq (capsule)	
	MGCD265			
mTOR	AP23573, MK8669	ridaforolimus (deforolimus)		rapalogue
	AY 22989, SILA 9268A, WY-090217	sirolimus	Rapamune	rapalogue
	AZD2014			
	AZD8055			TORC1/2, catalytic inhibitor
	BEZ235, NVP-BEZ235	dactolisib		catalytic, dual PI3K/mTOR inhibitor
	BGT226, NVP-BGT226			dual PI3K/mTOR inhibitor
	CC-223			
	CCI-779	temsirolimus	Torisel	rapalogue
	GDC-0980	apitolisib		catalytic, dual PI3K/mTOR inhibitor
	MLN0128, INK128			TORC1/2, catalytic inhibitor
	OSI-027			TORC1/2, catalytic inhibitor
	PF-4691502			catalytic, dual PI3K/mTOR inhibitor
	PI-103			catalytic, dual PI3K/mTOR inhibitor
	PKI-587, PF-05212384	gedatolisib		pan-PI3K inhibitor, catalytic, dual PI3K/mTOR inhibitor
	PWT33597			catalytic, dual PI3K/mTOR

	RAD001	everolimus	Afinitor	inhibitor rapalogue
	SF1126			catalytic, dual PI3K/mTOR inhibitor
	XL765, SAR245409	voxtalisib		catalytic, dual PI3K/mTOR inhibitor
NRAS	BGB-283			wild type-sparing NRAS inhibitor
NTRK1/2/3 (TRKA/B/C)	AZD7451			
	LOXO-101			
	MGCD516			
	PLX7486			
	RXDX-101	entrectinib		
NTRK1 (TRKA)	TSR-011			
	CEP-701	lestaurtinib		

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PDGFR	ABT-348	ilorasertib		
	AMG706	motesanib		
	BIBF1120	nintedanib	Vargatef	
	GW786034	pazopanib	Votrient	
	MGCD516			
	MLN518, CT53518	tandutinib		
	STI-571, CGP57148, CGP57148B	imatinib	Gleevec	
	X-82			inhibits all isoforms
PDGFR α	CP-868569	crenolanib		
	MP-470	amuvatinib		
	PKC412	midostaurin		
PDGFR β	ABT-869	linifanib		
	AG-013736	axitinib	Inlyta	
	BAY 43-9006	sorafenib	Nexavar	
	BAY 57-9352	telatinib		
	BAY 73-4506	regorafenib	Stivarga	
	SU-11248	sunitinib	Sutent	
	TSU 68, SU6668	orantinib		
	XL820			
PI3K	BAY 80-6946	copanisib		pan-PI3K inhibitor
	BEZ235, NVP-BEZ235	dactolisib		catalytic, dual PI3K/mTOR inhibitor
	BGT226, NVP-BGT226			dual PI3K/mTOR inhibitor
	BKM120	buparlisib		pan-PI3K inhibitor
	BYL719	alpelisib		
	GDC-0941	pictilisib		pan-PI3K inhibitor
	GDC-0980	apitolisib		catalytic, dual PI3K/mTOR inhibitor
	GSK2126458	omipalisib		pan-PI3K inhibitor
	GSK2636771			
	PF-4691502			catalytic, dual PI3K/mTOR inhibitor
	PI-103			catalytic, dual PI3K/mTOR inhibitor
	PKI-587, PF-05212384	gedatolisib		pan-PI3K inhibitor, catalytic
	PWT33597			catalytic, dual PI3K/mTOR inhibitor
	PX-866			
	SF1126			catalytic, dual PI3K/mTOR inhibitor
	XL147, SAR245408	pilaralisib		pan-PI3K inhibitor
	XL765, SAR245409	voxtalisib		catalytic, dual PI3K/mTOR inhibitor
		sophoretin, quercetin		
PI3K α	BYL719	alpelisib		
	GDC-0032	taselisib		
	INK-1117			
PI3K β/δ	AZD8186			
PI3K δ	CAL-101, GS-1101	idelalisib	Zydelig	
	TGR-1202			
PI3K δ/γ	IPI-145, INK-1197	duvelisib		
PKC		sophoretin, quercetin		
PKC α	LY317615	enzastaurin		
	PKC412	midostaurin		
PKC β	LY317615	enzastaurin		
PKC γ	LY317615	enzastaurin		
PKC ϵ	LY317615	enzastaurin		
PLK1	BI 6727	volasertib		
	BI 2536			
	GSK461364, GSK-461364			
	NMS-1286937			
	ON 01910.Na	rigosertib	Estybon	
RAF	BAY 43-9006	sorafenib	Nexavar	
	BAY 73-4506	regorafenib	Stivarga	
	LGX818	encorafenib		
	MLN2480			
	RO5126766, CHS126766			
	XL281, BMS-908662			
RET	AMG706	motesanib		
	AZD6474, ZD6474	vandetanib	Zactima, Caprelsa	
	BAY 43-9006	sorafenib	Nexavar	
	BAY 73-4506	regorafenib	Stivarga	
	MGCD516			
	MP-470	amuvatinib		
	SU-11248	sunitinib	Sutent	
	XL184	cabozantinib	Cabometyx (tablet), Cometriq (capsule)	
ROS1	PF-02341066	crizotinib	Xalkori	
	PF-06463922	lorlatinib		
	RXDX-101	entrectinib		

RPS6KB1 (p70S6K)	AT13148			
SCFR	AB1010	masitinib		
	MGCD516			
SRC	AZD0530	saracatinib		
	ABT-348	ilorasertib		
	BMS-354825	dasatinib	Sprycel	
	KX2-391			
	XL228			
SYK	GS-9973	entospletinib		
	PRT062070			
	R788	fostamatinib		
TBK1	CYT-387, CYT387	momelotinib		
TEK	MGCD265			
	XL184	cabozantinib	Cabometyx (tablet), Cometriq (capsule)	

TGFR1	LY2157299	galunisertib		
	TEW-7197			
TORC1/2	AZD8055			catalytic inhibitor
	MLN0128, INK128			catalytic inhibitor
	OSI-027			catalytic inhibitor
VEGFR	ABT-869	linifanib		
	ENMD-2076			
	EXEL-2880, GSK1363089, XL880	foretinib		
	HMPL-012	sulfatinib		
	PTK 787, CGP 79787, ZK 222584	vatalanib		
	TSU 68, SU6668	orantinib		
	X-82			inhibits all isoforms
	XL647, KD019			
VEGFR1/2/3	ABT-348	ilorasertib		
	AG-013736	axitinib	Inlyta	
	AMG706	motesanib		
	BAY 73-4506	regorafenib	Stivarga	
	BIBF1120	nintedanib	Vargatef	
	E-3810	lucitanib		
	GW786034	pazopanib	Votrient	
	HMPL-013	fruquintinib		
	KRN951, AV-951	tivozanib		
	MGCD265			
	MGCD516			
VEGFR2	AEE788, NVP-AEE788			
	AP24534	ponatinib	Iclusig	
	AZD2171	cediranib		
	AZD6474, ZD6474	vandetanib	Zactima, Caprelsa	
	BAY 43-9006	sorafenib	Nexavar	
	BMS-582664	brivanib		
	BMS-690514, EVRI			
	DP-1919.T0, DCC-2036	rebastinib tosylate		
	E7080	lenvatinib	Lenvima	
	PKC412	midostaurin		
	RAF265			
	SU-11248	sunitinib	Sutent	
	XL184	cabozantinib	Cabometyx (tablet), Cometriq (capsule)	
	XL820			
	YN968D1	apatinib		
VEGFR2/3	BAY 57-9352	telatinib		
WEE1	AZD1775, MK1775			

治疗性抗体				
靶点	研发代码	通用名	商品名	备注
ANGPT1/2 (ANG1/2)	AMG 780			
ANGPT2 (ANG2)	MEDI-3617			
	REGN910	nesvacumab		
	RO5520985, RG7221	vanucizumab		
B4GALNT1	Ch14.18, ch14.18-UTC	dinutuximab	Unituxin	
BAFF		belimumab	Benlysta	
CCR4	KW-0761	mogamulizumab		
CD19	MEDI-551	inebilizumab		
CD19/CD3	MT103, MT-103, MEDI-538	blinatumomab	Blincyto	
CD20	GA101	obinutuzumab	Gazyvaro	
	IDEC-102, IDEC-C2B8	rituximab	Rituxan, Mabthera	
	IDEC Y2B8	ibritumomab tiuxetan	Zevalin	drug can be bound via tiuxetan to ¹¹¹ In or ⁹⁰ Y
	TG-1101, LFB-R603	ublitzumab		
		ofatumumab	Arzerra, HuMax-CD20	
CD22	CAT-8015	moxetumomab pasudotox		
CD27	CDX-1127	varlilumab		
CD33		gemtuzumab	Mylotarg	
CD37	TRU-016	otlertuzumab		
CD38		daratumumab	Darzalex, HuMax-CD38	
	SAR650984			
CD52	Campath	alemtuzumab		
CD70	ARGX-110			
	SGN-70, h1F6	vorsetuzumab		
CD74	hLL1	milatuzumab		
CD105	TRC105			
CD248 (TEM1)	MORAb-004	ontuxizumab		
CSF1R	RG7155, RO5509554	emactuzumab		
CSF2	KB003	lenzilumab		
CTLA4	CP675,206	tremelimumab		
	MDX-010, MDX-101, BMS-734016	ipilimumab	Yervoy	
DLL4	OMP-21M18	dencizumab		
EGFL7	MEGF0444A	parsatuzumab		
EGFR (HER1/ERBB1)	ABT-806, mAb-806	depatuxizumab		
	ABX-EGF	panitumumab	Vectibix	
	h-R3, TheraCIM	nimotuzumab		
	IMC-11F8	necitumumab	Portrazza	
	IMC-C225	cetuximab	Erbix	
	MEHD7945 A	duligotuzumab		
	MM-151			
	RO5083945, GA201	imgatuzumab		
	Sym004	futuximab		
FZD7	OMP-18R5	vantictumab		
HER2 (ERBB2/neu)	GT-Mab 7.3-GEX	glycooptimized trastuzumab-GEX		
	MGAH22	margetuximab		
	MM-111			
	MOAB 2C4	pertuzumab	Perjeta, Omnitarg	
		trastuzumab	Herceptin	
HER3 (ERBB3)	AV 203			
	LJM716			
	MEHD7945 A	duligotuzumab		
	MM-111			
	MM-121, SAR256212	seribantumab		
	MM-141			
	REGN1400			
	U3-1287, AMG 888	patritumab		
HGFR	AMG 102	rilotumumab		
	AV-299, SCH900105	ficlatuzumab		
IGF-1R	AMG-479	ganitumab		
	BI 836845			
	BIIB022			
	CP-751871	figitumumab		
	IMC-A12	cixutumumab		
	MEDI-573	dusigitumab		
	MK-0646	dalotuzumab		
	MM-141			
IGF-2R	BI 836845			
	MEDI-573			
KIR	BMS-986015	lirilumab		
LOXL2	AB0024, GS-6624	simtuzumab		
MET	ABT-700			
	LY2875358	emibetuzumab		
	MetMab, RO5490258, OA-5D	onartuzumab		
MMP9	GS-5745			

NOTCH1	OMP-52M51	brontictuzumab		
NRPI	MNRP1685A	vesencumab		
PD-1	AMP-224			
	BMS-936558, MDX-1106, ONO-4538	nivolumab	Opdivo	
	CT-011	pidilizumab		
	MK-3475, lambrolizumab	pembrolizumab	Keytruda	
PD-L1	MDX-1105, BMS-936559			
	MEDI-4736	durvalumab		
	MPDL3280A, RG7446	atezolizumab		
	MSB0010718C			
PDGFRα	IMC-3G3	olatumab		
	MEDI-575	tovetumab		
PIGF		ziv-aflibercept	Zaltrap	

SEMA4D	VX15/2503			
SLAMF7 (CS1/CD319/CRACC)	HuLuc63	elotuzumab	Empliciti	
TGFBR2	IMC-TR1			
TNFRSF9 (CD137/4-1BB)	BMS 663513	urelumab		
	PF-05082566			
TRAIL-R1/DR4,R2/D5	AMG951, rhApo2L/TRAIL	dulanermin		
TRAIL-R1/D4	HGS-ETR1	mapatumumab		
TRAIL-R2/D5	AMG-655	conatumumab		
	HGS-ETR2	lexatumumab		
	LBY135			
	PRO95780	apomab		
VEGFA		bevacizumab	Avastin	
	RO5520985, RG7221	vanucizumab		
		ziv-aflibercept	Zaltrap	
VEGFB		ziv-aflibercept	Zaltrap	
VEGFC	VGX-100			
VEGFR2	IMC-1121B	ramucirumab	Cyramza	

抗体偶联药物ADC					
靶点	偶联药物	研发代码	通用名	商品名	备注
CD19	maytansinoid DM4	SAR3419			
	monomethyl auristatin F	SGN-CD19A, SGN-19A	denintuzumab mafodotin		
CD22	calicheamycin	CMC-544	inotuzumab ozogamicin		
	monomethyl auristatin E	RG7593, DCDT2980S	pinatuzumab vedotin		
CD30	monomethyl auristatin E	SGN-35	brentuximab vedotin	Adcetris	
CD33	calicheamycin		gemtuzumab ozogamicin	Mylotarg	
CD56	maytansinoid DM1	IMGN-901	lorvotuzumab mertansine		
CD70	Prodrug of duocarmycin analogue	BMS-936561, MDX-1203			
	monomethyl auristatin F	SGN-75	vorsetuzumab mafodotin		
CD79b	monomethyl auristatin E	RG7596, DCDS4501A	polatuzumab vedotin		
CEACAM5	SN-38	IMMU-130			
EGFR	monomethyl auristatin F	ABT-414	depatuxizumab mafodotin		
EGFRvIII	maytansinoid DM1	AMG 595			
FGFR2	novel auristatin cytotoxic agent	BAY 1187982			
FOLH1 (PSMA)	monomethyl auristatin E	PSMA ADC			
FOLR1	maytansinoid DM4	IMGN853	mirvetuximab soravtansine		
GPNCMB	monomethyl auristatin E	CDX-011	glembatumumab vedotin		
GUCY2C (GCC)	monomethyl auristatin E	MLN0264			
HER2 (ERBB 2/neu)	maytansinoid DM1	PRO132365	ado trastuzumab emtansine	Kadcyla	
Mesothelin	monomethyl auristatin E	DMOT4039A			
MUC16 protein (CA-125)	monomethyl auristatin E	Anti-MUC16 ADC, DMUC5754A	sofituzumab vedotin		
Nectin-4	monomethyl auristatin E	AGS-22M6E	enfortumab vedotin		
SLC34A2 (NaPi2b)	monomethyl auristatin E	DNIB0600A, RG7599	lifastuzumab vedotin		
SLITRK6 (SLIT and NTRK-like protein 6)	monomethyl auristatin E	AGS15E			
STEAP1	monomethyl auristatin E	DSTP3086S, RG7450			
TACSTD2 (TROP2)	SN-38	IMMU-132	sacituzumab govitecan		

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免疫疗法				
靶点	研发代码	通用名	商品名	备注
CD19	CTL019, CART19	tisagenlecleucel-T		
CD227 (MUC1)	BLP25	tecemotide		
	IMMU-107, hPAM4, IgG-DOTA	clivatuzumab tetraxetan		
CTLA4	CP675,206	tremelimumab		
	MDX-010, MDX-101, BMS-734016	ipilimumab	Yervoy	
IDO1	INCB024360	epacadostat		
	NLG-919			
IL-6		siltuximab	Sylvant	
IL-6R	R-1569	tocilizumab	Actemra	
IL-1 β	ACZ885	canakinumab	Ilaris	
KIR	BMS-986015	lirilumab		
PD-1	AMP-224			
	BMS-936558, MDX-1106, ONO-4538	nivolumab	Opdivo	
	CT-011	pidilizumab		
	MK-3475, lambrolizumab	pembrolizumab	Keytruda	
PD-L1	MDX-1105, BMS-936559			
	MEDI-4736	durvalumab		
	MPDL3280A, RG7446	atezolizumab		
	MSB0010718C			
SEMA4D	VX15/2503			
TNFRSF9 (CD137/4-1BB)	BMS 663513	urelumab		
	PF-05082566			
Other	Ad.p53 DC vaccine			cancer vaccine
	AGS-003	rocapuldencel-T		cancer vaccine
	APC8015	sipuleucel-T	Provenge	cancer vaccine
	BPX-201			cancer vaccine
	CDX-1401			
	CG8123, GVAX			cancer vaccine
	DC-TC	eltrapuldencel-T		
	DCVAC/Pca			cancer vaccine
	E75 Peptide	nelipepimut-S		
	ETBX-011			cancer vaccine
	GI-6301			cancer vaccine
	HyperAcute-Lung, HAL	tergenpumatucl-L		
	HS-110	viagenpumatucl-L		cancer vaccine
	HS-410	vesigenurtacel-L		cancer vaccine
	IMA901			cancer vaccine
	MAGE-A3			cancer vaccine
	NLG-12036, HyperAcute-Melanoma	dorgenmeltucel-L		cancer vaccine
	OncoVex GM-CSF, T-VEC	talimogene laherparepvec	Imlygic	cancer vaccine
	ONT-10			cancer vaccine
	polyvalent melanoma vaccine, PMCV			cancer vaccine
	Prostvac-V	rilimogene galvacirepvec		
	PSA/IL-2/GM-CSF vaccine			cancer vaccine
	TG4010, MVA-MUC1-IL2			cancer vaccine
	TheraCys	BCG vaccine		cancer vaccine
	TSF	aldesleukin	Proleukin	
	VGX-3100			cancer

	VAX-3100			vaccine
		belagenpumatucel-L	Lucanix	cancer vaccine
		enadenotucirev		
		interferon alpha-2b	Intron A	
		peginterferon alfa-2b	PEG-Intron	

其他 靶点	研发代码	通用名	商品名	备注
activin	ACE-011	sotatercept		fusion protein
ADORA3 (A3AR)	CF102			
ANGPT1/2 (ANG1/2)	AMG 386	trebananib		
Arginine	ADI-PEG 20	pegargiminase		
Bcl-2	ABT-263, ABT-737	navitoclax		
	AT101			
	GDC-0199, ABT-0199, ABT-199, RG7601	venetoclax	Venclexta	
	GX15-070	obatoclax		
	G3139	oblimerson	Genasense	
BRD2/3/4	OTX015			
CPAA	NEO-102, NPC-1C	ensituximab		
CXCR4	BMS-936564, MDX-1338	ulocuplumab		
	LY2510924			
Cyclin D	ON 014185, ON 01305	briciclib		
	ON 013105	briciclib sodium		
DOT1L	EPZ-5676			
EC1456				Small molecule-drug conjugate
endo-Beta-D-glucuronidase heparanase	PI-88	muparfostat		
EZH2	EPZ-6438, E7438			
Farnesyl proteintransferase (FTase)	IND 58359	tipifarnib	Zarnestra	
FGF 1/2	PI-88	muparfostat		
FGFR1	HGS1036, FP-1039, GSK3052230			
Folate receptor	EC145	vintafolide		folate-vinca alkaloid conjugate
	Gamma-secretase	MK0752		
FOLH1 (PMSA)	G-202	mipsagargin		
FZD8	OMP-54F28	ipafricept		
HDAC	4SC-201	resminostat		
	CRA-024781, PCI-24781	abexinostat		
	FK228	romidepsin	Istodax	
	LBH589	panobinostat	Farydak, Faridak	
	MGCD0103	mocetinostat		
	MK-0683	vorinostat	Zolinza	
	MS-275, SNDX-275	entinostat		
	PXD101	belinostat	Beleodaq	
	SHP-141	reminostat		
HDAC6	ACY-1215, ACY-63	ricolinostat		
HDM2/P53	SAR405838, MI-773			
HSP27	OGX-427	apatorsen		
Hedgehog	PF-04449913	glasdegib		
HSP90	I7-AAG, KOS-953	tanespimycin		
	AT13387	onalespib		
	AUY922	luminespib		
	Debio 0932			
	IPI-504	retaspimycin		
	SNX-5422			
	STA-9090	ganetespib		
Inhibitor of Apoptosis Protein (IAP)	XL888			
IDH1	TL32711	birinapant		
IDO	AG-120			
IDO	1-methyl-d-tryptophan	indoximod		
KDM1A (LSD1)	GSK2879552			
KIF11 (KSP/EG5)	ARRY-520	filanesib		
	4SC-205			
MDM2	RG7388, RO503781	idasanutlin		
Mesothelin	MORAB-009	amatuximab		
Microtubule	BAL101553			
Nedd8 Activating Enzyme (NAE)	MLN4924	pevonedistat		
Notch Pathway	MK0752			
	OMP-59R5	tarextumab		
	PF-03084014	nirogacestat		
	REGN421			
	RO4929097, R4733			
PARP	ABT-888	veliparib		
	AG-014699, PF-01367338	rucaparib		
	AZD2281	olaparib	Lynparza	
	BMN 673	talazoparib		
	BSI-201	iniparib		
Phosphotyrosine	MK4827	niraparib		
Phosphotyrosine		bavituximab		
PMSA	BIND-014			
PORCN	LGR974			

PP2A	LB-100			
PPAR- γ	CS 7017, RS5444			
Proteasome	LDP 341, MLN341, PS-341	bortezomib	Velcade	
	MLN9708	ixazomib	Ninlaro	
	PR-171	carfilzomib	Kyprolis	
PTCH	GDC-0449	vismodegib	Erivedge	
Smoothered	GDC-0449	vismodegib	Erivedge	
	IPI-926, sandegib	patidegib		
	LDE225, erismodegib	sonidegib	Odomzo	
	LY2940680	taladegib		
STAT3	OPB-31121			
TGF- β	ACE-536	luspatercept		fusion protein
TLR8	VTX-2337, VTX-378	motolimod		
TLR5	CBLB502	entolimod		

VEGF	PI-88	muparfostat		
XPO1 (CRM1)	KPT-330	selinexor		
Other	BB1503			
	BB1608			
	CC-4047	pomalidomide	Pomalyst	thalidomide derivative
	CC-5013	lenalidomide	Revlimid	thalidomide analog
		thalidomide	Thalomid	
	M402	necuparanib		
	NHS-IL12			
	OBI-822, OPT822	adagloxad simolenin		
	PRX302	topsalsyn		
	T-VEC	talimogene laherparepvec	Imlygic	
	TH-302	evofosfamide		
	VAL-083, NSC-1323313	dianhydrogalactitol		
	VB-111	remetinostat		